Andreas PLENGE

Decision Systems Engineer

Quantitative and analytical engineer with strong expertise in stochastic modeling, data-driven optimization, and scalable algorithmic systems. Skilled in Python, SQL, and probabilistic modeling, with experience in real-time control, forecasting, and data governance. Passionate about leveraging mathematical and software expertise to solve complex problems in performance sensitive systems.

Contact Information

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Languages

- Danish — Native
- English — Fluent

Core Expertise

- Optimization Under Uncertainty
- Real-Time Algorithmic Control
- Distributed Energy Market Systems
- Probabilistic Graph Algorithms
- Performance-Critical Software Architecture

Mathematical Foundation

- PDE Theory
- Stochastic Calculus
- Dynamical Systems
- Probability & Statistical Modeling

Implementation Stack

- Python
- TypeScript
- SQL / NoSQL
- TensorFlow / PyTorch
- Kubernetes / Docker
- MQTT & Event-Driven Systems

References are available upon request

Work Experience

Bodil Energi — August 2024 – Present

Tech Lead | Energy Systems Engineer — prev. ML Engineer

Designed and developed the core grid stabilization algorithm, orchestrating real-time coordination of distributed household devices using scalable combinatorial methods and linear-complexity optimizations. Conducted code reviews and collaborated with the engineering team to ensure system robustness and maintainability.

Deployed the platform across households, using consumption forecasts to participate in ancillary service markets and controlling devices based on live frequency monitoring, enabling reliable and efficient stabilization of the electricity grid.

Since mid-2025, serving as the primary in-house engineer responsible for the platform.
◇ TypeScript, Python, SQL / NoSQL, Kubernetes / Docker, Optimization Under Uncertainty, Real-Time Algorithmic Control, Distributed Energy Market Systems, Performance-Critical Software Architecture, Dynamical Systems, Probability & Statistical Modeling

KOMBIT — August 2022 – August 2024

IT Consultant | Power BI Developer — prev. Student Developer

Designed dashboards for complex data visualization. Automated data workflows, data quality controls and administrative tasks, to enhance efficiency. Provided technical support, data analysis, and ad hoc problem-solving.
◇ Python, R, Power BI (DAX / M), Data Quality & Compliance

Education

MSc in **Software Design** — 2024

at IT Univeristy of Copenhagen

- Specialization in **machine learning**
- **Thesis:** Proposed a machine learning algorithm for the k-approximate nearest neighbors problem, applying a hierarchical structure to inverted file indexes using k-means clustering.

MSc in **Mathematics** — 2020

at University of Copenhagen

- Specialization in **dynamical systems**
- **Thesis:** unfinished

BSc in **Mathematics** — 2018

at University of Copenhagen

- Specialization in **mathematical finance**
- **Thesis:** Explored the connection between the heat equation and Brownian motion, combining advanced concepts in probability and differential equations.